

Drake Lyrics Analysis

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Intro

I started with a dataset from [**Kaggle**](#) containing all of Drake's lyrics at the time of its creation. In this project, I perform textual normalization on song lyrics, evaluate word frequency distributions, analyze key terms across albums, and investigate the relationship between lyric themes (specifically the word "love") and overall track popularity.

```
#cleaning up the data
drake_combined=c()
for(i in 1:11){
  d=read.csv(paste0("drake_data_",i,".csv"))
  drake_combined=rbind(drake_combined,d)
}
drake_combined$lyrics=str_to_lower(drake_combined$lyrics)
drake_combined=mutate(drake_combined,lyrics = gsub("[&\\\\\\,\\.\\\\?\\\\![:digit:]]\\\\\\\\n", "",lyrics))
```

Top 10 words

This table displays the top 10 most frequently occurring words.

```
lyrics_only=drake_combined$lyrics
lyrics_words=str_split(lyrics_only," ", simplify=TRUE)
lyrics_words_vec=as.vector(lyrics_words)
lyrics_words_vec=lyrics_words_vec[lyrics_words_vec!=""]
my.table=as.data.frame(table(lyrics_words_vec))
my.sort=arrange(my.table,desc(Freq))
colnames(my.sort)=c("Word","Frequency")
kbl(my.sort[1:15,])
```

Word	Frequency
the	4402
i	3971
you	3951
to	2838
a	2540
it	1934
and	1759
my	1733
me	1722
that	1606
in	1423
on	1320
i'm	1286
like	1226
know	1183

However, the words are mostly filler words, so here I cleaned up the data a bit to make a more informative table.

```
freq_lyrics <- lyrics_words_vec[!lyrics_words_vec %in% stop_words$word]
freq_table <- as.data.frame(table(freq_lyrics))
freq_sorted <- arrange(freq_table, desc(Freq))
colnames(freq_sorted)=c("Word", "Frequency")
kbl(freq_sorted[1:15,])
```

Word	Frequency
love	448
shit	427
wanna	369
yeah	301
niggas	297
girl	292
time	275
nigga	265
gotta	248
fuck	233
money	218
'em	177
drake	173
feel	172
real	164

Now, a word cloud. I used [this](#) very helpful tutorial!

```
wordcloud(freq_sorted$Word, freq_sorted$Frequency, scale=c(5,0.5)
, 100
, random.order=FALSE
, rot.per=0.35
, use.r.layout=FALSE, colors=brewer.pal(8, "Dark2"))
```

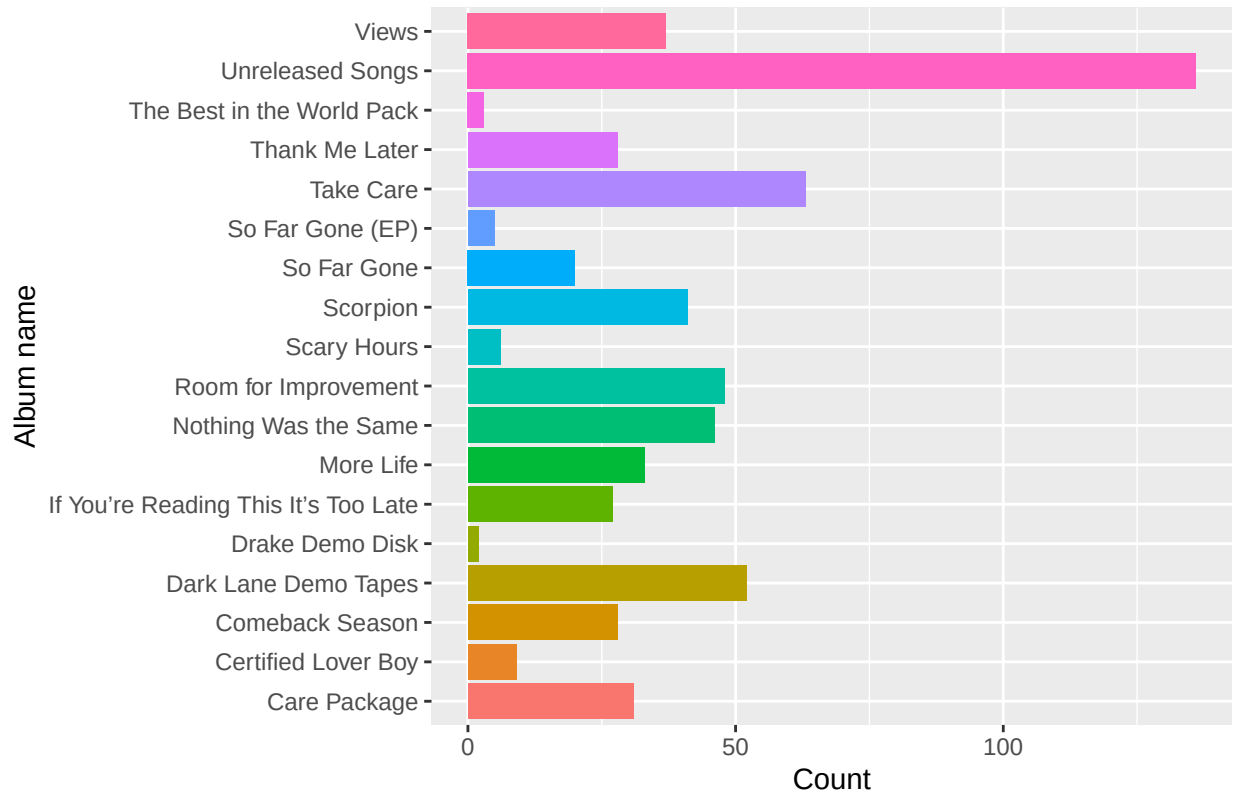

my.albums	love	drake	respect	money	trust
If You're Reading This It's Too Late	27	32	0	11	3
Unreleased Songs	136	71	6	17	5
Room for Improvement	48	79	6	39	1
So Far Gone	20	37	0	32	0
More Life	33	63	1	6	12
Take Care	63	55	1	38	4
Nothing Was the Same	46	42	2	24	6
Comeback Season	28	67	3	35	2
Care Package	31	31	0	18	12
Views	37	47	1	9	7
Scorpion	41	89	7	10	2
The Best in the World Pack	3	5	0	8	0
Certified Lover Boy	9	19	0	5	0
Thank Me Later	28	41	1	27	2
Dark Lane Demo Tapes	52	43	4	5	12
Drake Demo Disk	2	7	1	5	0
Scary Hours	6	0	0	1	0
So Far Gone (EP)	5	7	0	19	0

```

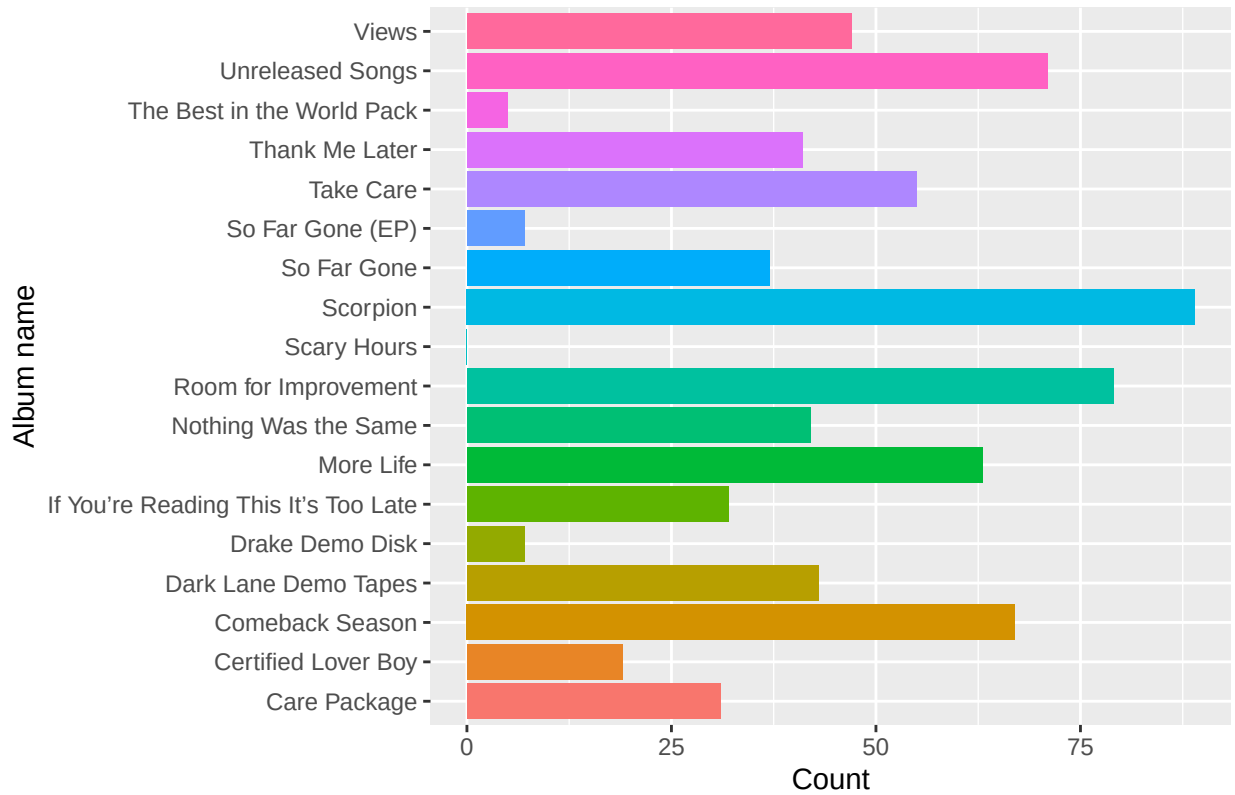
table_long=pivot_longer(table,cols=my.words,names_to="word",values_to="count")
for(i in 1:5){
  sub=table_long[table_long$word==my.words[i],]
  myplot=ggplot(sub)+geom_bar(aes(x=my.albums,y=count,fill=my.albums),stat="identity") + labs(title=pas
  print(myplot)
}

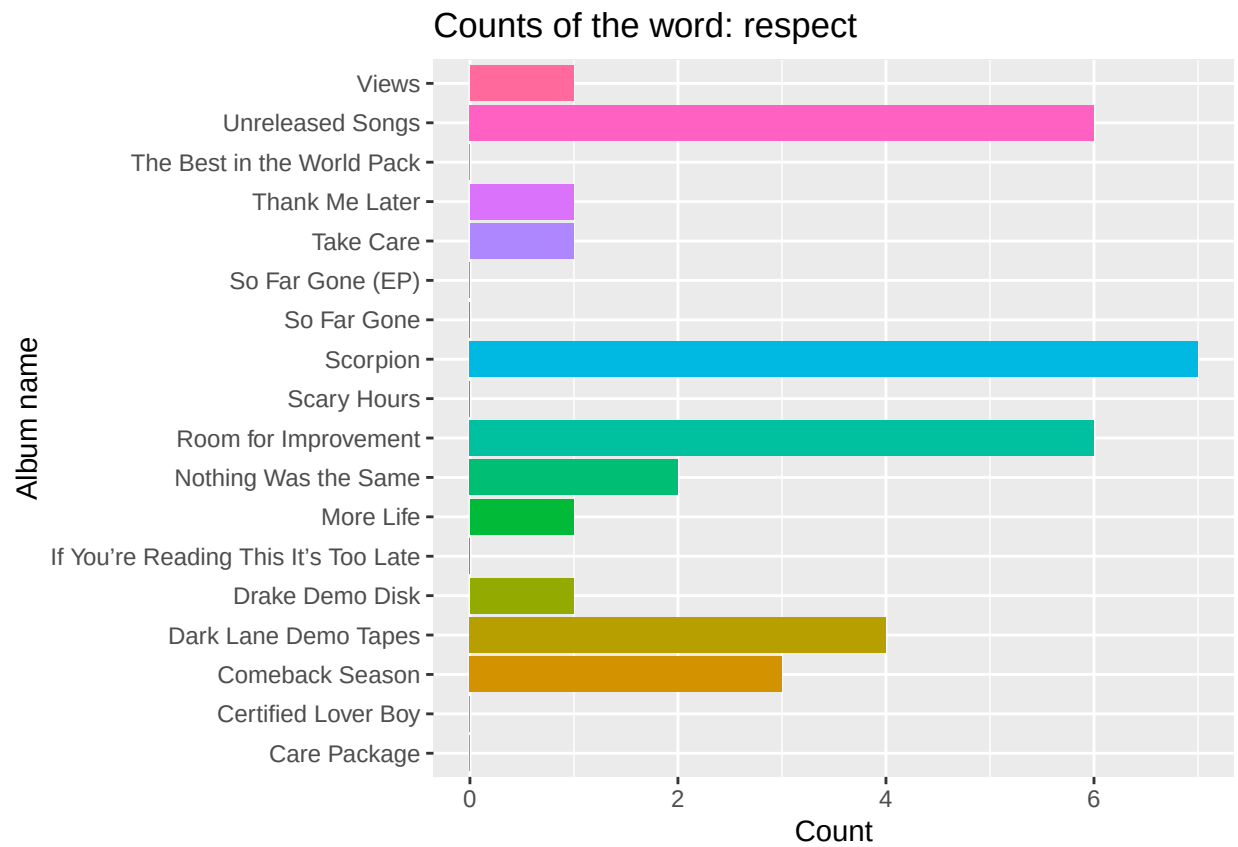
```

Counts of the word: love

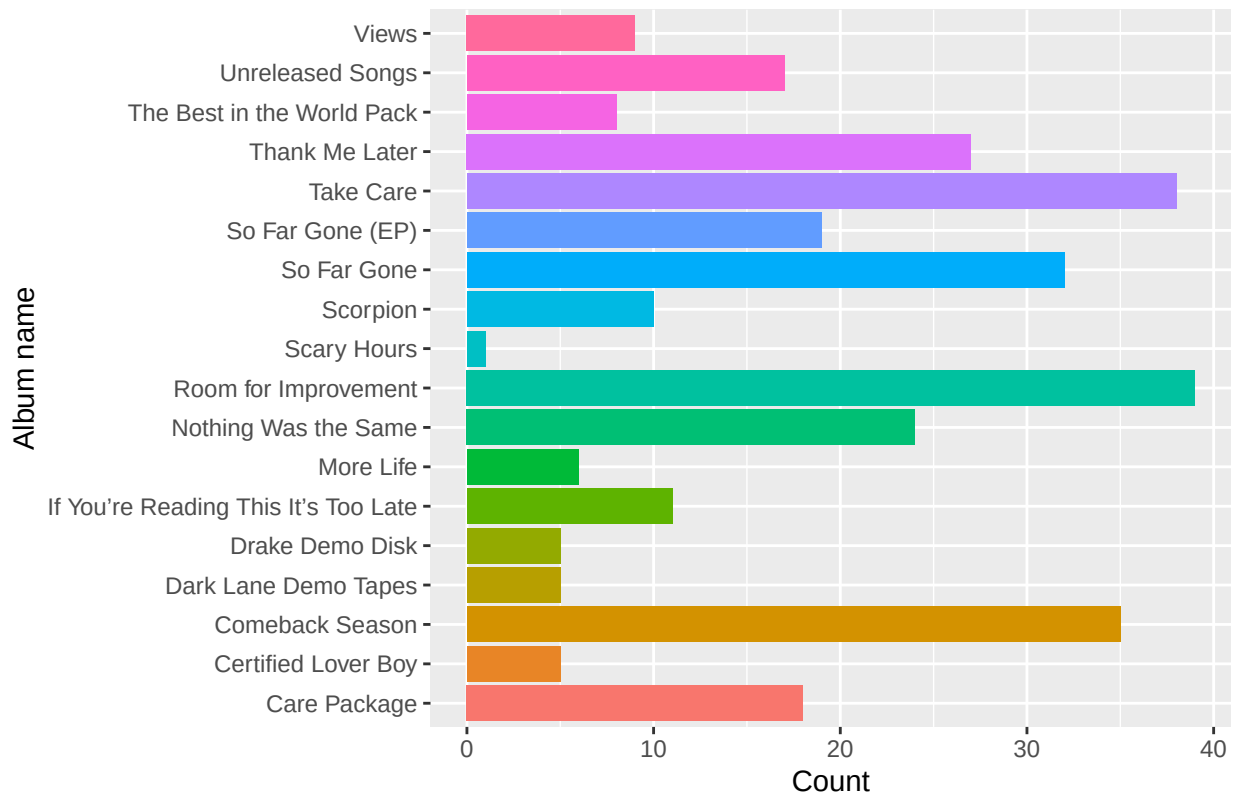


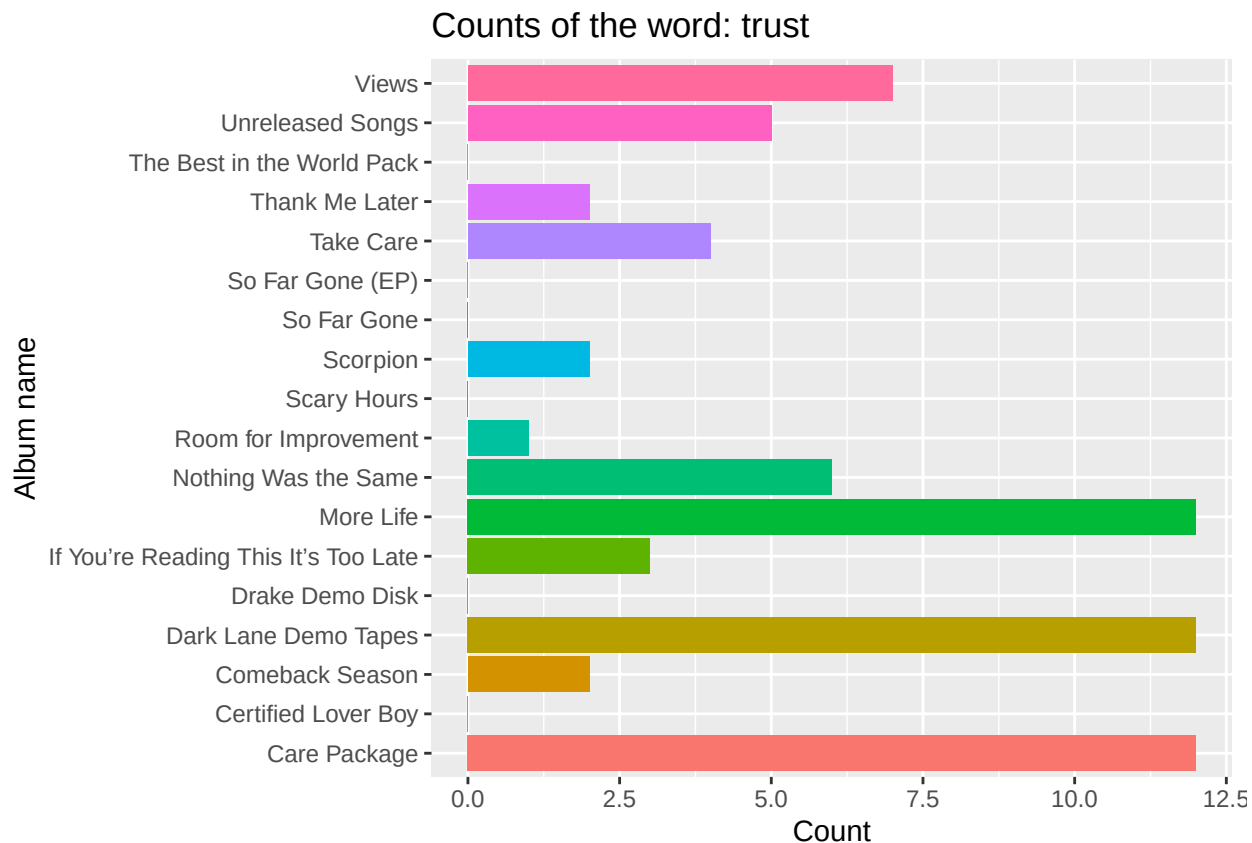
Counts of the word: drake





Counts of the word: money

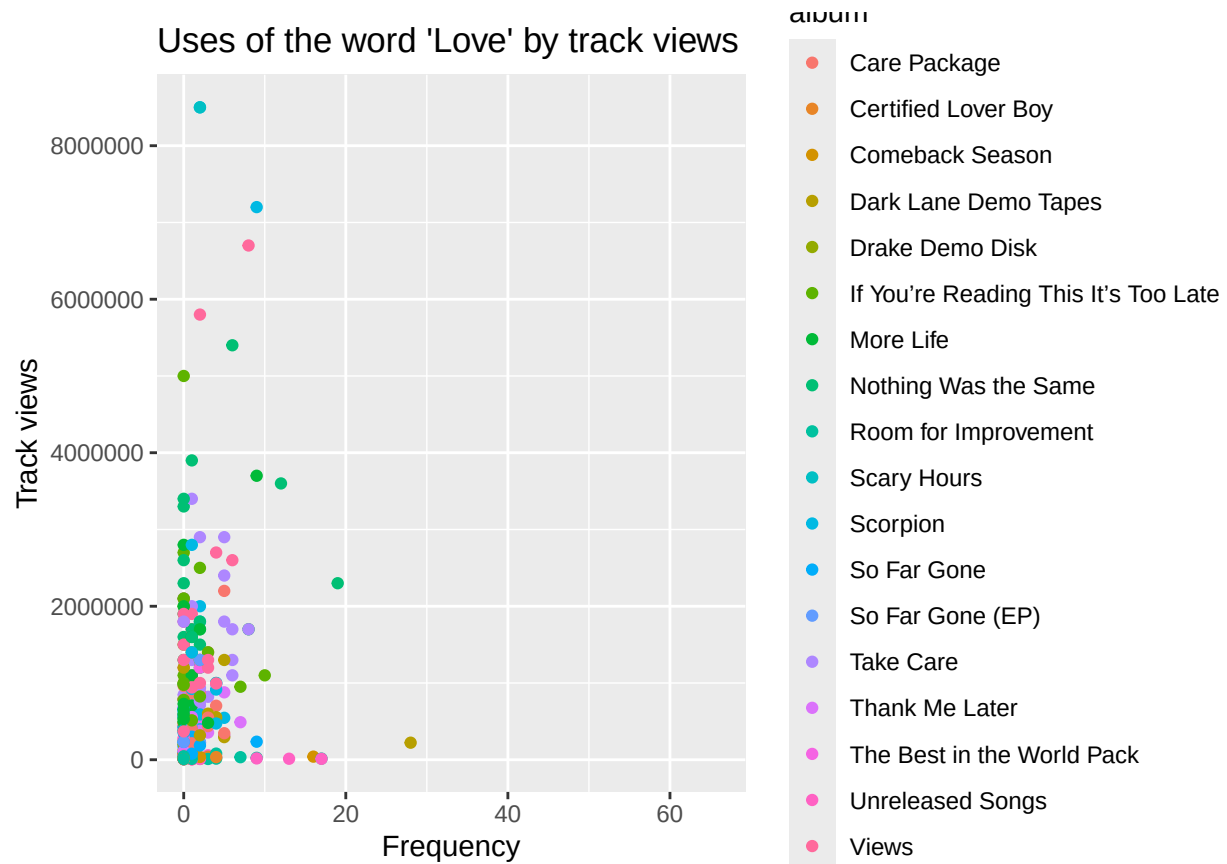




Here I track the success of tracks by use of the word “love”

```
drake_combined$love=str_count(drake_combined$lyrics,"love")
view_number=c()
for(i in 1:length(drake_combined$track_views)){
  if(is.na(drake_combined$track_views[i])){
    view_number=c(view_number,NA)
  }
  else if(str_detect(drake_combined$track_views[i],"K")==TRUE)
  {
    view_number[i]=parse_number(drake_combined$track_views[i])*1000
  }
  else if(str_detect(drake_combined$track_views[i],"M")==TRUE)
  {
    view_number[i]=parse_number(drake_combined$track_views[i])*1000000
  }
  else
  {
    view_number[i]=parse_number(drake_combined$track_views)
  }
}
drake_combined$track_views=view_number
ggplot(drake_combined) + geom_point(aes(x=love,y=track_views,color=album)) + labs(title="Uses of the word love")
```

```
## Warning: Removed 17 rows containing missing values or values outside the scale range ('geom_point()')
```



We can see that, in general, his tracks that did the best didn't use the word "love" too much.

Conclusion/Reflections

We definitely saw some pretty interesting results overall. We saw that tracks with high view counts (> 50M views) usually feature lower frequencies of the word "love". Also, the most frequently used words were a bit surprising to me, I was definitely not expecting "love" to be the top word!